Real-Time Traffic Signal Control Simulator Using QNX RTOS

Real-time embedded system leveraging QNX RTOS for intelligent traffic management

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Introduction

Traffic signal control is an essential part of modern transportation systems, where accurate timing and coordination are required to ensure smooth and safe traffic flow. Efficient signal management plays a key role in reducing congestion and improving road safety.

Problem Statement

However, most traditional traffic signal systems operate on fixed timing, which leads to congestion, inefficiency, and delays under varying traffic conditions. This highlights the need for a system that can ensure real-time responsiveness and precise timing control.

Project Objectives

Real-Time Control

Implement precise signal timing using QNX RTOS shared memory communicationwithin the simulation.

Safety Compliance

Develop fail-safe simulation logic to validate redundant state management and signal safety.

State Machine Logic

Implement a deterministic Finite State Machine (FSM) to govern transition states in software simulation.

Multithreading Simulation

Utilize concurrent threads to simulate multi-lane traffic flow and signal processing asynchronously.

Synchronization Mechanisms

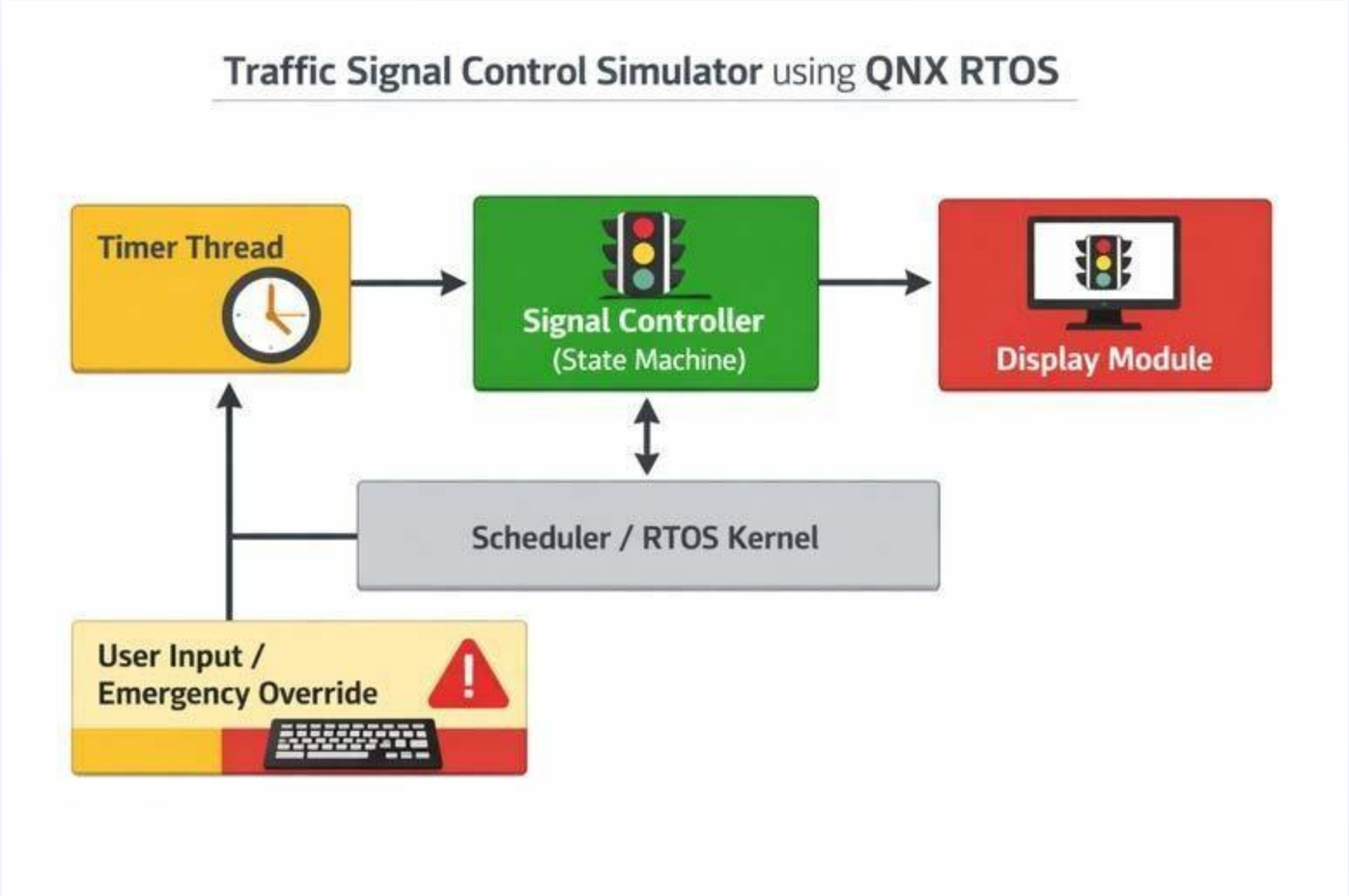
Apply mutexes and semaphores to ensure robust data integrity between traffic sensor data streams.

Emergency Override

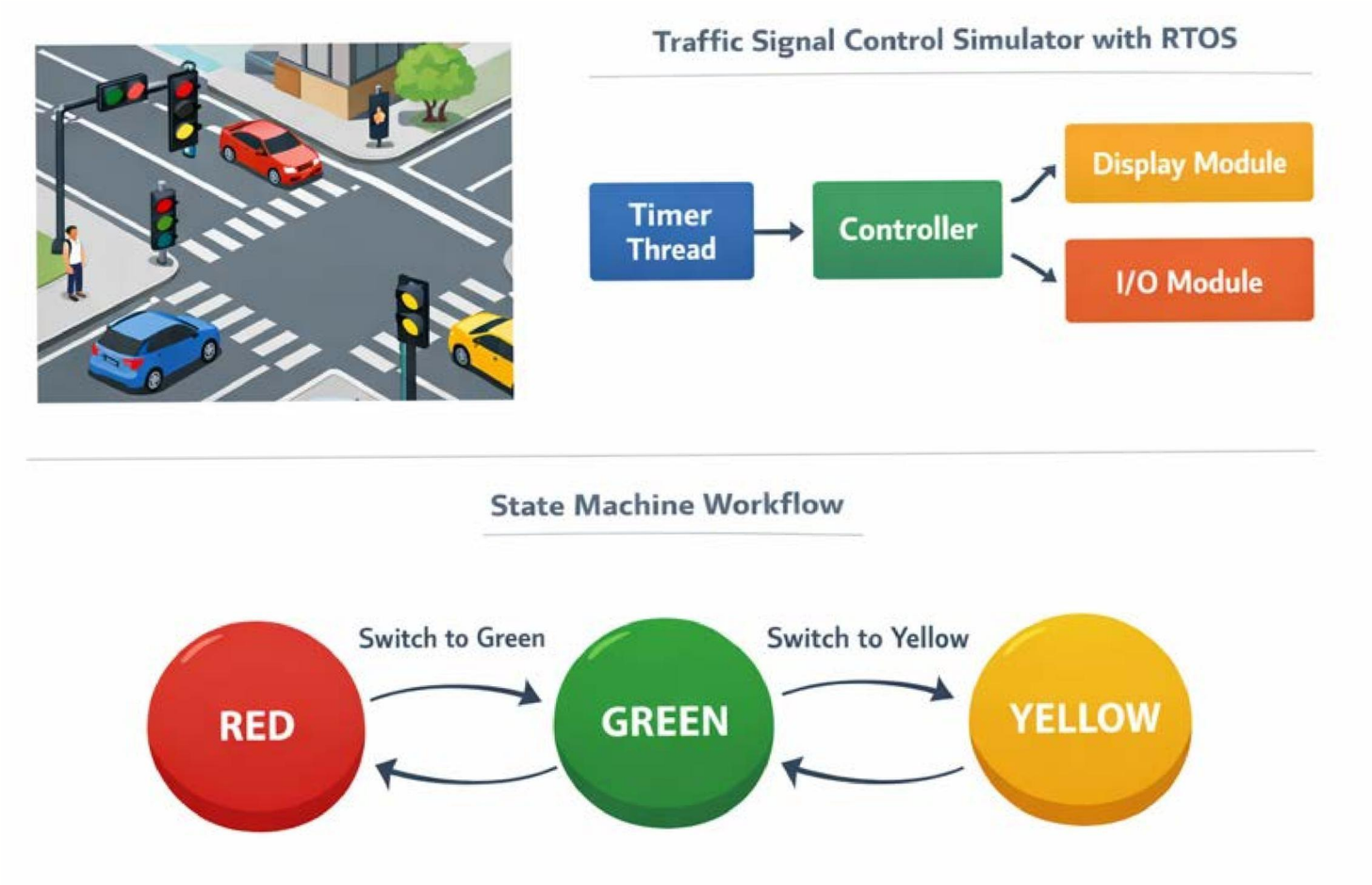
Develop prioritized software interrupts to handle emergency vehicle overrides within the simulation loop.

System Block Diagram

- **Timer Thread**
  Generates time intervals to control how long each traffic signal stays ON (red, yellow, green).
- **Signal Controller (State Machine)**
  Decides which signal should be active based on timing logic and traffic sequence.
- **Display Module**
  Shows the current traffic signal status (Red / Yellow / Green).
- **User Input / Emergency Override**
  Allows manual control in emergency situations (like ambulance priority).
- **Scheduler / RTOS Kernel**
  Manages all threads using priority-based scheduling to ensure real-time response.



System Implementation

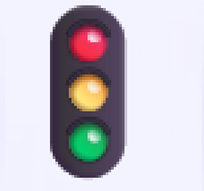


Implementation Components

Traffic Sensors (Simulated)

Provides input conditions (traffic density)

QNX Kernel
Handles thread scheduling using preemptive priority



Signal Outputs
Controls traffic lights (LED/display simulation)

CLI Interface
Allows user to start/stop simulation using commands

Workflow Process

Initialisation Phase

System starts, QNX RTOS boots, and all components (threads, timers, I/O) are configured. Default traffic signal state is set.

Data Acquisition

Traffic input is *simulated* (vehicle density, timing conditions) and sent to the controller using shared memory communication.

Analysis & Decision

Controller processes inputs using a **state machine algorithm** and decides signal timing dynamically.

Signal Execution

Traffic lights change (Red → Green → Yellow) based on computed timing with **safe delays and synchronization**.

Solution Overview

QNX RTOS Core

Ensures **deterministic real-time execution** using priority-based preemptive scheduling.

Controls timing of traffic signals with high precision and reliability.

Shared Memory Mechanism

Different modules communicate using **shared memory (for signal state communication)** ensuring fast and reliable data exchange.

Fault Recovery

A **watchdog timer** monitors system health.

If any module fails or delays, system switches to **safe mode (default signal behavior)**.

Logging & Monitoring

Console-based logging of signal transitions (RED → GREEN → YELLOW) for **debugging and system tracking.**

Advantages & Future Scope

Key Advantages

- Reduced average wait time through adaptive timing
- Hard real-time guarantees prevent deadline misses
- Modular design enables easy feature expansion
- POSIX-compliant APIs ensure portability

Future Enhancements

- Machine learning integration for traffic pattern prediction
- IoT connectivity for city-wide coordination
- Pedestrian detection with priority handling
- Emergency vehicle pre-emption systems

Output (After Execution)

```
Problems  Tasks  Console ×  Properties  Ta
bin.Traffic [C/C++ QNX Application]
Press s to start:
s

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COMMAND LIST
c1 → Only 2-Road
c2 → Only 3-Road
c3 → Both

2-ROAD:
eA eB → Emergency
dA dB → Density
n2     → Normal

3-ROAD:
e1 e2 e3 → Emergency
d1 d2 d3 → Density
n3       → Normal

q → Quit
--------------------------------
```

```
c1
Select mode (c1/c2/c3):
[2-ROAD] A:GREEN B:RED

[2-ROAD] A:GREEN B:RED

[2-ROAD] A:GREEN B:RED

[2-ROAD] A:GREEN B:RED

[2-ROAD] A:GREEN B:RED

[2-ROAD] A:YELLOW B:RED

[2-ROAD] A:YELLOW B:RED

[2-ROAD] A:RED B:GREEN

[2-ROAD] A:RED B:GREEN

[2-ROAD] A:RED B:GREEN

[2-ROAD] A:RED B:GREEN

[2-ROAD] A:RED B:GREEN

[2-ROAD] A:RED B:GREEN

[2-ROAD] A:RED B:YELLOW
```

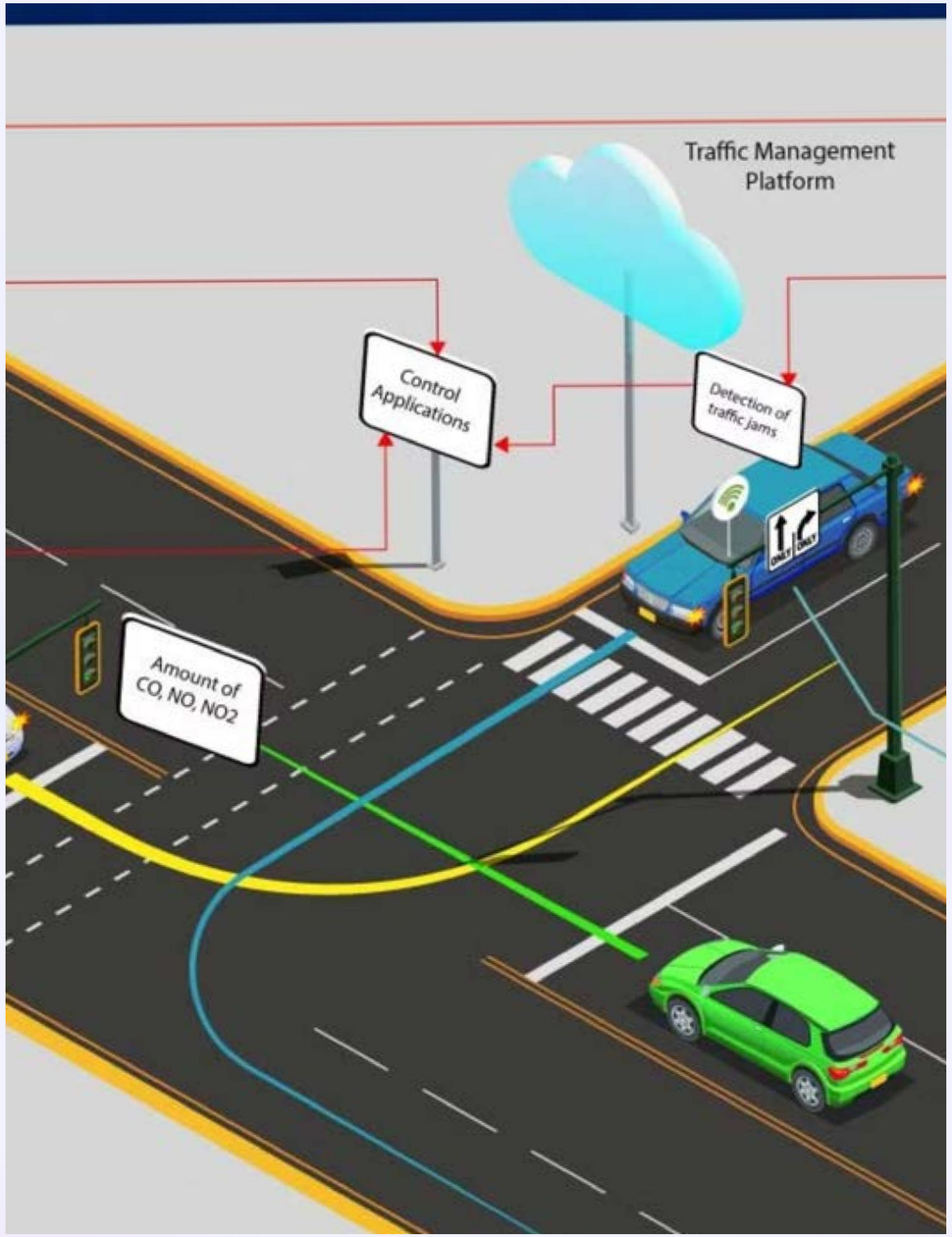
```
[2-ROAD][EMERGENCY] A:GREEN B:RED
eB

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN

[2-ROAD][EMERGENCY] A:RED B:GREEN
```

```
c3
[2-ROAD] A:GREEN B:RED

[2-ROAD] A:GREEN B:RED

[2-ROAD] A:GREEN B:RED


[2-ROAD] A:GREEN B:RED

[3-ROAD] R1:GREEN R2:RED R3:RED

[3-ROAD] R1:GREEN R2:RED R3:RED

[2-ROAD] A:GREEN B:RED

[3-ROAD] R1:GREEN R2:RED R3:RED

[2-ROAD] A:GREEN B:RED

[3-ROAD] R1:GREEN R2:RED R3:RED

[2-ROAD] A:GREEN B:RED
```

```
c2
Select mode (c1/c2/c3):
[3-ROAD] R1:GREEN R2:RED R3:RED

[3-ROAD] R1:GREEN R2:RED R3:RED

[3-ROAD] R1:GREEN R2:RED R3:RED

[3-ROAD] R1:GREEN R2:RED R3:RED

[3-ROAD] R1:GREEN R2:RED R3:RED

[3-ROAD] R1:YELLOW R2:RED R3:RED

[3-ROAD] R1:YELLOW R2:RED R3:RED

[3-ROAD] R1:RED R2:GREEN R3:RED

[3-ROAD] R1:RED R2:GREEN R3:RED

[3-ROAD] R1:RED R2:GREEN R3:RED

[3-ROAD] R1:RED R2:GREEN R3:RED

[3-ROAD] R1:RED R2:GREEN R3:RED
```

```
[3-ROAD] R1:RED R2:YELLOW R3:RED
d3

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN

[3-ROAD][DENSITY] R1:RED R2:RED R3:GREEN
```

```
e2

[3-ROAD][EMERGENCY] R1:RED R2:GREEN R3:RED

[3-ROAD][EMERGENCY] R1:RED R2:GREEN R3:RED

[3-ROAD][EMERGENCY] R1:RED R2:GREEN R3:RED

[3-ROAD][EMERGENCY] R1:RED R2:GREEN R3:RED

[3-ROAD][EMERGENCY] R1:RED R2:GREEN R3:RED

[3-ROAD][EMERGENCY] R1:RED R2:GREEN R3:RED

[3-ROAD][EMERGENCY] R1:RED R2:GREEN R3:RED
q
[3-ROAD][EMERGENCY] R1:RED R2:GREEN R3:RED

Simulation Ended
```



Conclusion

Technical Achievement

Successfully implements real-time traffic management system demonstrating QNX RTOS capabilities in embedded applications.

Practical Impact

Our solution will address real-world congestion challenges with measurable efficiency improvements.

THANK YOU

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